

ITNET04 Network Design Project

Phase 1

Alonzo IT Training Center Network Documentation

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1. IP Addressing Scheme

1.1 Main Office

1.1.1 Subnet Table

Subnetwork	VLAN	Network Address	Subnet Mask	Host Address Range	Notes
172.16.0.0/22					
Main_Students	100	172.16.0.0	255.255.255.128	172.16.0.1 - 172.16.0.126	120/126 hosts
Main_Guests	200	172.16.0.128	255.255.255.192	172.16.0.129 - 172.16.0.190	56/62 hosts
Main_Instructors	300	172.16.0.192	255.255.255.224	172.16.0.193 172.16.0.222	26/30 hosts
Main_Operations	400	172.16.0.224	255.255.255.224	172.16.0.225 172.16.0.254	27/30 hosts
Main_Management	500	172.16.1.0	255.255.255.224	172.16.1.1 - 172.16.1.30	2+/30 hosts
Main_IT	600	172.16.1.32	255.255.255.240	172.16.1.33 - 172.16.1.46	10/14 hosts
Main_Services	700	172.16.1.48	255.255.255.240	172.16.1.49 - 172.16.1.62	4/14 hosts

1.1.2 Host Address Assignment Table

Subnet / VLAN Name: Main_Students

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0/0.100	172.16.0.1	255.255.255.128	N/A
Main_RTR2 G0/0/0.100	172.16.0.2	255.255.255.128	N/A
Virtual Gateway (Group 100)	172.16.0.3	255.255.255.128	N/A
Training Room 1			
Main_TR1-Student-PC1 - Main_TR1-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
Training Room 2			
Main_TR2-Student-PC1 - Main_TR2-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
Training Room 3			
Main_TR3-Student-PC1 - Main_TR3-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
Training Room 4			
Main_TR4-Student-PC1 - Main_TR4-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
Training Room 5			
Main_TR5-Student-PC1 - Main_TR5-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
Training Room 6			

Main_TR6-Student-PC1 - Main_TR6-Student-PC20	DHCP (172.16.0.4 - 126)	255.255.255.128	172.16.0.3
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Subnet / VLAN Name: Main_Guests

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0/0.200	172.16.0.129	255.255.255.192	N/A
Main_RTR2 G0/0/0.200	172.16.0.130	255.255.255.192	N/A
Virtual Gateway (Group 200)	172.16.0.131	255.255.255.192	N/A
Library			
Main_Library-Guest-PC - Main_Library-Guest-PC6	DHCP (172.16.0.132 - 190)	255.255.255.192	172.16.0.131

Subnet / VLAN Name: Main_Instructors

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0.300	172.16.0.193	255.255.255.224	N/A
Main_RTR2 G0/0.300	172.16.0.194	255.255.255.224	N/A
Virtual Gateway (Group 300)	172.16.0.195	255.255.255.224	N/A
Faculty			
Main_Instructor-PC1 - Main_Instructor-PC10	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR1-Instructor-P C1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR2-Instructor-P C1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR3-Faculty-PC1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR4-Instructor-P C1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR5-Instructor-P C1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195
Main_TR6-Instructor-P C1	DHCP (172.16.0.196 - 222)	255.255.255.224	172.16.0.195

Subnet / VLAN Name: Main_Operations

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0/0.400	172.16.0.225	255.255.255.224	N/A
Main_RTR2 G0/0/0.400	172.16.0.226	255.255.255.224	N/A
Virtual Gateway (Group 400)	172.16.0.227	255.255.255.224	N/A
Library			
Main_Librarian-PC1	DHCP (172.16.0.228 - 254)	255.255.255.224	172.16.0.227
Faculty			

Main_Secretary-PC1	DHCP (172.16.0.228 - 254)	255.255.255.224	172.16.0.227
Administrative Office			
Main_Operations-PC1 - Main_Operations-PC11	DHCP (172.16.0.228 - 254)	255.255.255.224	172.16.0.227

Subnet / VLAN Name: Main_Management

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0/0.500	172.16.1.1	255.255.255.224	N/A
Main_RTR2 G0/0/0.500	172.16.1.2	255.255.255.224	N/A
Virtual Gateway (Group 500)	172.16.1.3	255.255.255.224	N/A
Aries-Server1	172.16.1.4	255.255.255.224	172.16.1.3
Main_Management-PC	172.16.1.29	255.255.255.224	172.16.1.3
Switch Management Interface			
Main_Dist-SW1	172.16.1.5	255.255.255.224	172.16.1.3
Main_Dist-SW2	172.16.1.6	255.255.255.224	172.16.1.3
Main_IT-Acc-SW	172.16.1.7	255.255.255.224	172.16.1.3
Main_Admin-Acc-SW	172.16.1.8	255.255.255.224	172.16.1.3
Main_Faculty-Acc-SW	172.16.1.9	255.255.255.224	172.16.1.3
Main_Library-Acc-SW	172.16.1.10	255.255.255.224	172.16.1.3
Main_TR1-Acc-SW	172.16.1.11	255.255.255.224	172.16.1.3
Main_TR2-Acc-SW	172.16.1.12	255.255.255.224	172.16.1.3
Main_TR3-Acc-SW	172.16.1.13	255.255.255.224	172.16.1.3
Main_TR4-Acc-SW	172.16.1.14	255.255.255.224	172.16.1.3
Main_TR5-Acc-SW	172.16.1.15	255.255.255.224	172.16.1.3
Main_TR6-Acc-SW	172.16.1.16	255.255.255.224	172.16.1.3
Wireless			
WLC	172.16.1.17	255.255.255.224	172.16.1.3
Main_First-Floor-AP	DHCP (172.16.1.18 - 30)	255.255.255.224	172.16.1.3
Main_Second-Floor-AP	DHCP (172.16.1.18 - 30)	255.255.255.224	172.16.1.3

Subnet / VLAN Name: Main_IT

Device / Host	IP Address	Subnet Mask	Default Gateway
Main_RTR1 G0/0/0.600	172.16.1.33	255.255.255.240	N/A
Main_RTR2 G0/0/0.600	172.16.1.34	255.255.255.240	N/A
Virtual Gateway (Group 600)	172.16.1.35	255.255.255.240	N/A
IT Office			
Main_IT-PC1 - Main_IT-PC5	DHCP (172.16.1.36 - 46)	255.255.255.240	172.16.1.35

Subnet / VLAN Name: Main_Services

Device / Host	IP Address	Subnet Mask	Default Gateway
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Main_RTR1 G0/0/0.700	172.16.1.49	255.255.255.240	N/A
Main_RTR2 G0/0/0.700	172.16.1.50	255.255.255.240	N/A
Virtual Gateway (Group 700)	172.16.1.51	255.255.255.240	N/A
Orion-Server1	172.16.1.52	255.255.255.240	172.16.1.51
Aquarius Server1	172.16.1.53	255.255.255.240	172.16.1.51
Main_Service-Printer1	DHCP (172.16.1.54 - 62)	255.255.255.240	172.16.1.51
Main_Service-Printer2	DHCP (172.16.1.54 - 62)	255.255.255.240	172.16.1.51

1.2 Quezon City Branch

1.2.1 Subnet Table

Subnetwork	VLAN	Network Address	Subnet Mask	Host Address Range	Notes
192.168.0.0/23					
QC_Students	100	192.168.0.0	255.255.255.128	192.168.0.1 - 192.168.0.126	80/126 hosts
QC_Management	500	192.168.0.128	255.255.255.224	192.168.0.129 - 192.168.0.158	2+/30 hosts
QC_Operations	400	192.168.0.160	255.255.255.240	192.168.0.161 - 192.168.0.174	10/14 hosts
QC_Instructors	300	192.168.0.176	255.255.255.240	192.168.0.177 - 192.168.0.190	8/14 hosts
QC_IT	600	192.168.0.192	255.255.255.240	192.168.0.193 - 192.168.0.206	6/14 hosts

1.2.2 Host Address Assignment Table

Subnet / VLAN Name: QC_Students

Device / Host	IP Address	Subnet Mask	Default Gateway
QC_RTR1 G0/0/0.100	192.168.0.1	255.255.255.128	N/A
<i>Training Room 1</i>			
QC_TR1-Student-PC1 - QC_TR1-Student-PC20	DHCP (192.168.0.2 - 126)	255.255.255.128	192.168.0.1
<i>Training Room 2</i>			
QC_TR2-Student-PC1 - QC_TR2-Student-PC20	DHCP (192.168.0.2 - 126)	255.255.255.128	192.168.0.1
<i>Training Room 3</i>			
QC_TR3-Student-PC1 - QC_TR3-Student-PC20	DHCP (192.168.0.2 - 126)	255.255.255.128	192.168.0.1
<i>Training Room 4</i>			
QC_TR4-Student-PC1 - QC_TR4-Student-PC20	DHCP (192.168.0.2 - 126)	255.255.255.128	192.168.0.1

Subnet / VLAN Name: QC Management

Device / Host	IP Address	Subnet Mask	Default Gateway
QC_RTR1 G0/0/0.500	192.168.0.129	255.255.255.224	N/A
Leo-Server1	192.168.0.130	255.255.255.224	192.168.0.129
QC_Management-PC	192.168.0.131	255.255.255.224	192.168.0.129
QC_Admin-Acc-SW	192.168.0.132	255.255.255.224	192.168.0.129
QC_Faculty-Acc-SW	192.168.0.133	255.255.255.224	192.168.0.129
QC_IT-Acc-SW	192.168.0.134	255.255.255.224	192.168.0.129
QC_TR1-Acc-SW	192.168.0.135	255.255.255.224	192.168.0.129
QC_TR2-Acc-SW	192.168.0.136	255.255.255.224	192.168.0.129
QC_TR3-Acc-SW	192.168.0.137	255.255.255.224	192.168.0.129
QC_TR4-Acc-SW	192.168.0.138	255.255.255.224	192.168.0.129

Subnet / VLAN Name: QC Operations

Device / Host	IP Address	Subnet Mask	Default Gateway
QC_RTR1 G0/0/0.400	192.168.0.161	255.255.255.240	N/A
QC_Operations-PC1 - QC_Operations-PC10	DHCP (192.168.0.162 - 174)	255.255.255.240	192.168.0.161

Subnet / VLAN Name: QC Instructors

Device / Host	IP Address	Subnet Mask	Default Gateway
QC_RTR1 G0/0/0.300	192.168.0.177	255.255.255.240	N/A
QC_Instructor-PC1 - QC_Instructor-PC4	DHCP (192.168.0.178 - 190)	255.255.255.240	192.168.0.177
QC_TR1-Instructor-PC1	DHCP (192.168.0.178 - 190)	255.255.255.240	192.168.0.177
QC_TR2-Instructor-PC1	DHCP (192.168.0.178 - 190)	255.255.255.240	192.168.0.177
QC_TR3-Instructor-PC1	DHCP (192.168.0.178 - 190)	255.255.255.240	192.168.0.177
QC_TR4-Instructor-PC1	DHCP (192.168.0.178 - 190)	255.255.255.240	192.168.0.177

Subnet / VLAN Name: QC IT

Device / Host	IP Address	Subnet Mask	Default Gateway
QC_RTR1 G0/0/0.600	192.168.0.193	255.255.255.240	N/A
QC_IT-PC1 - QC_IT-PC6	DHCP (192.168.0.194 - 206)	255.255.255.240	192.168.0.193

1.3 Annex Branch**1.3.1 Subnet Table**

Subnetwork	VLAN	Network Address	Subnet Mask	Host Address Range	Notes
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192.168.2.0/23					
<i>Annex_Students</i>	<i>100</i>	<i>192.168.2.0</i>	<i>255.255.255.128</i>	<i>192.168.2.1 - 192.168.2.126</i>	<i>60/126 hosts</i>
<i>Annex_Management</i>	<i>500</i>	<i>192.168.2.128</i>	<i>255.255.255.224</i>	<i>192.168.2.129 - 192.168.2.158</i>	<i>2+/30 hosts</i>
<i>Annex_Instructors</i>	<i>300</i>	<i>192.168.2.160</i>	<i>255.255.255.240</i>	<i>192.168.2.161 - 192.168.2.174</i>	<i>6/14 hosts</i>
<i>Annex_IT</i>	<i>600</i>	<i>192.168.2.176</i>	<i>255.255.255.240</i>	<i>192.168.2.177 - 192.168.2.190</i>	<i>10/14 hosts</i>

1.3.2 Host Address Assignment Table

Subnet / VLAN Name: *Annex_Students*

Device / Host	IP Address	Subnet Mask	Default Gateway
Device / Host	IP Address	Subnet Mask	Default Gateway
<i>Annex_RTR1 G0/0/0.100</i>	<i>192.168.2.1</i>	<i>255.255.255.128</i>	<i>N/A</i>
<i>Training Room 1</i>			
<i>Annex_TR1-Student-PC1 - Annex_TR1-Student-PC20</i>	<i>DHCP (192.168.2.2 - 126)</i>	<i>255.255.255.128</i>	<i>192.168.2.1</i>
<i>Training Room 2</i>			
<i>Annex_TR2-Student-PC1 - Annex_TR2-Student-PC20</i>	<i>DHCP (192.168.2.2 - 126)</i>	<i>255.255.255.128</i>	<i>192.168.2.1</i>
<i>Training Room 3</i>			
<i>Annex_TR3-Student-PC1 - Annex_TR3-Student-PC20</i>	<i>DHCP (192.168.2.2 - 126)</i>	<i>255.255.255.128</i>	<i>192.168.2.1</i>

Subnet / VLAN Name: *Annex_Management*

Device / Host	IP Address	Subnet Mask	Default Gateway
<i>Annex_RTR1 G0/0/0.500</i>	<i>192.168.2.129</i>	<i>255.255.255.224</i>	<i>N/A</i>
<i>Annex_Capricorn-Server 1</i>	<i>192.168.2.130</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Main_Management-PC</i>	<i>192.168.2.131</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Annex_Faculty-Acc-SW</i>	<i>192.168.2.132</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Annex_IT-Acc-SW</i>	<i>192.168.2.133</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Annex_TR1-Acc-SW</i>	<i>192.168.2.134</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Annex_TR2-Acc-SW</i>	<i>192.168.2.135</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>
<i>Annex_TR3-Acc-SW</i>	<i>192.168.2.136</i>	<i>255.255.255.224</i>	<i>192.168.2.129</i>

Subnet / VLAN Name: *Annex_Instructors*

Device / Host	IP Address	Subnet Mask	Default Gateway
<i>Annex_RTR1 G0/0/0.300</i>	<i>192.168.2.161</i>	<i>255.255.255.240</i>	<i>N/A</i>
<i>Annex_Instructor-PC1 - Annex_Instructor-PC3</i>	<i>DHCP (192.168.2.162 - 174)</i>	<i>255.255.255.240</i>	<i>192.168.2.161</i>
<i>Annex_TR1-Instructor-PC1</i>	<i>DHCP (192.168.2.162 - 174)</i>	<i>255.255.255.240</i>	<i>192.168.2.161</i>
<i>Annex_TR2-Instructor-PC1</i>	<i>DHCP</i>	<i>255.255.255.240</i>	<i>192.168.2.161</i>

	(192.168.2.162 - 174)		
Annex_TR3-Instructor-PC1	DHCP (192.168.2.162 - 174)	255.255.255.240	192.168.2.161

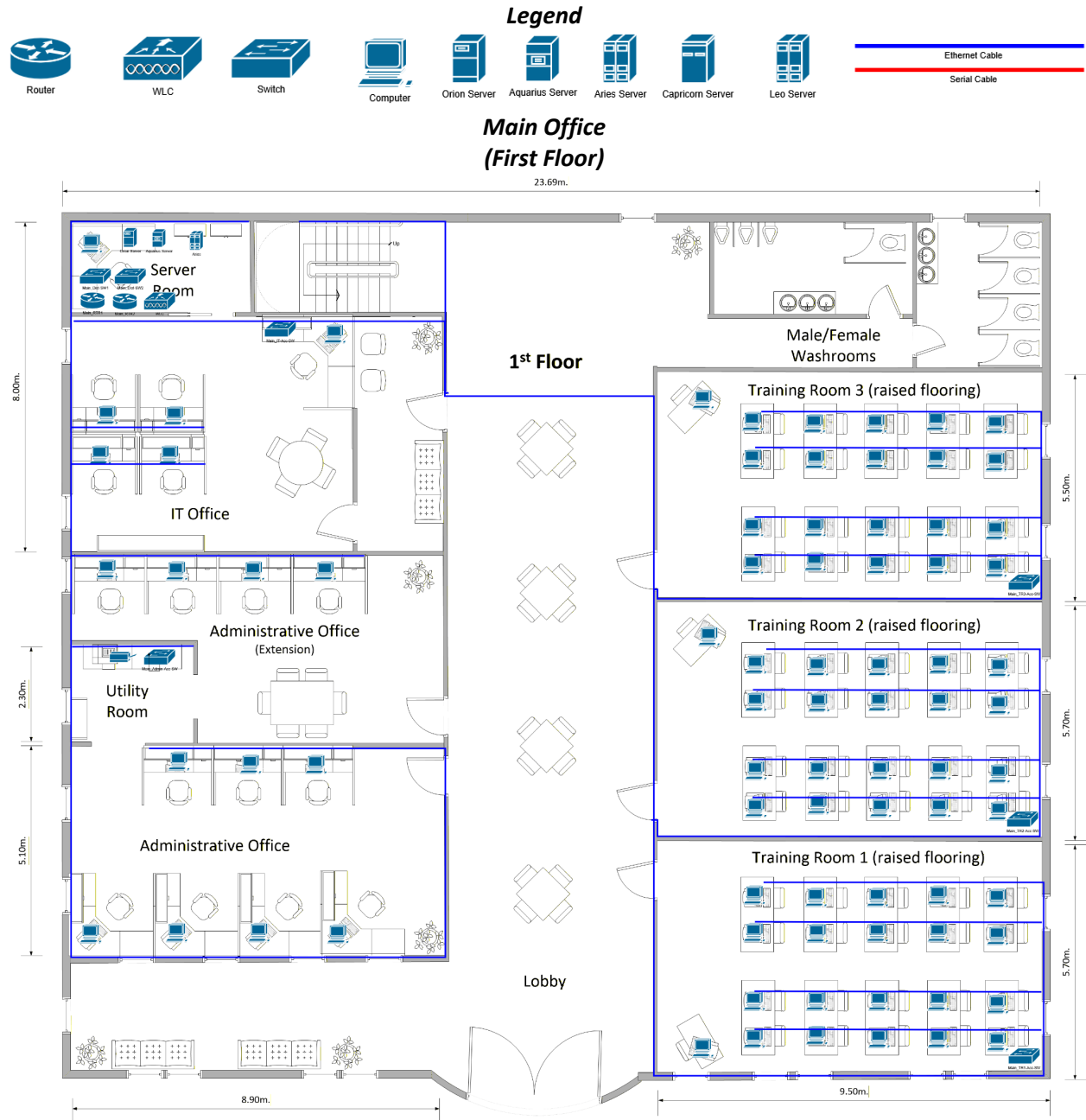
Subnet / VLAN Name: Annex_IT

Device / Host	IP Address	Subnet Mask	Default Gateway
Annex_RTR1 G0/0/0.600	192.168.2.177	255.255.255.240	N/A
QC_IT-PC1	DHCP (192.168.2.178 - 190)	255.255.255.240	192.168.2.177
QC_IT-PC2	DHCP (192.168.2.178 - 190)	255.255.255.240	192.168.2.177

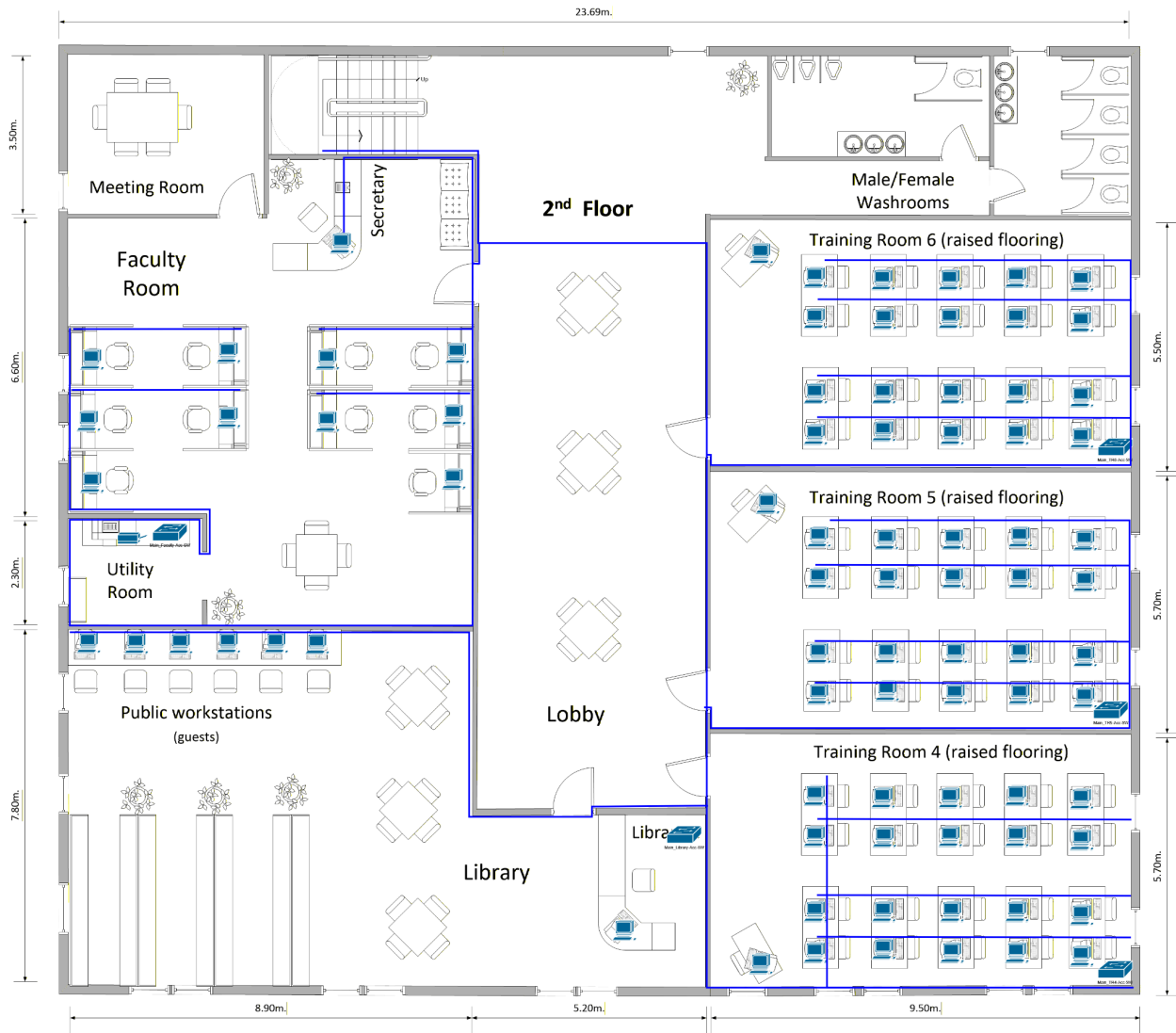
2. Physical Topology

2.1. Main Office

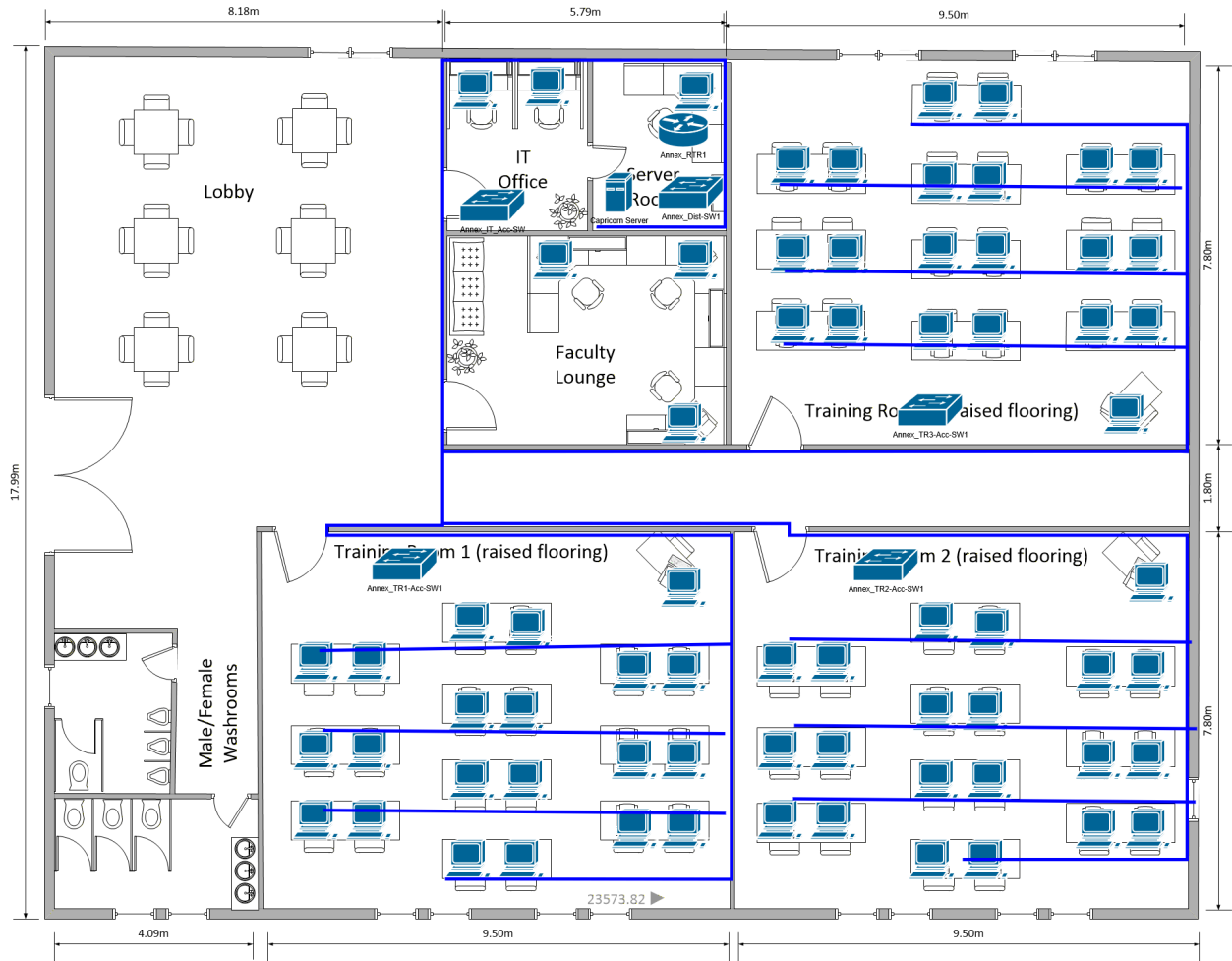
2.1.1 Floor Plan

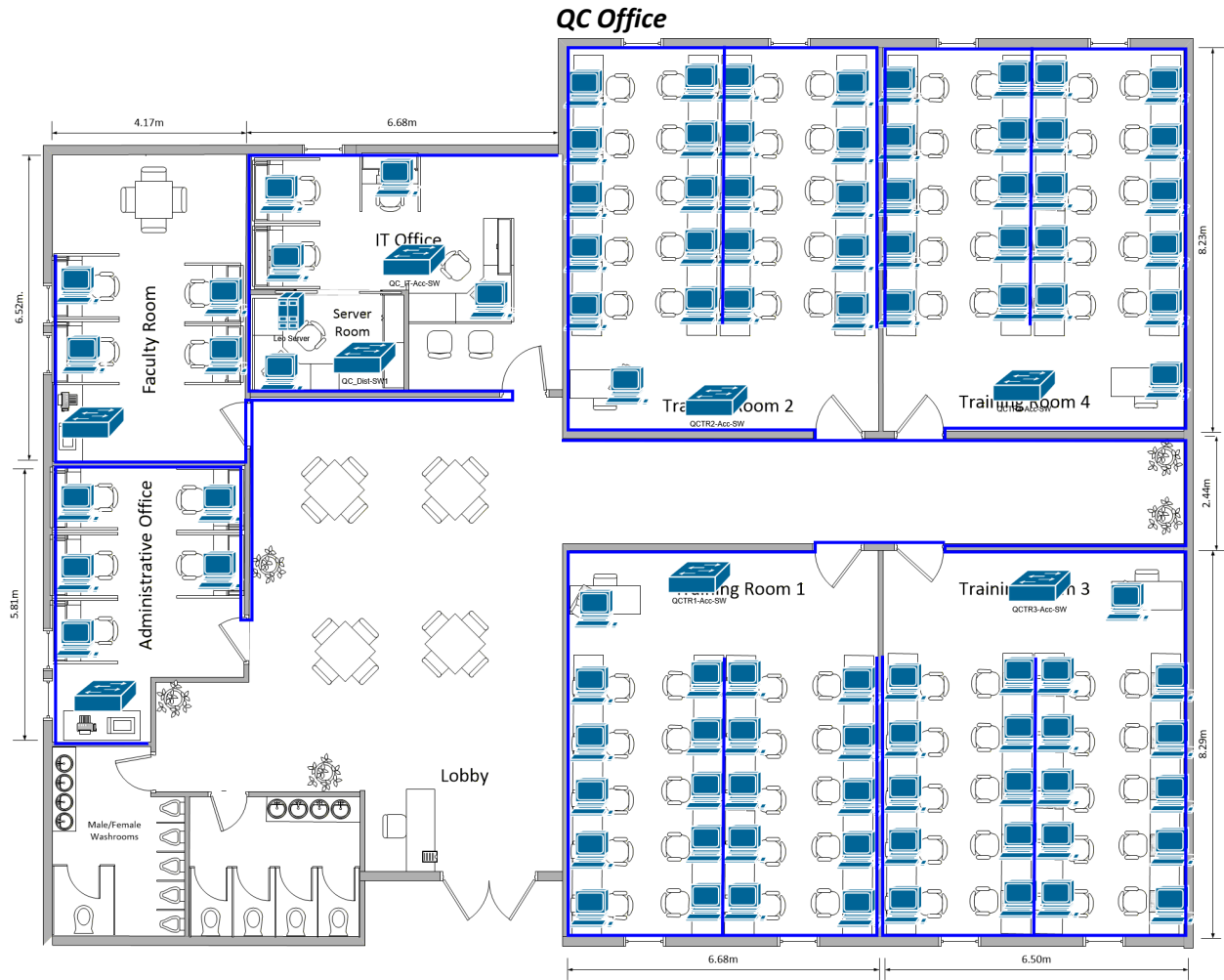


**Main Office
(Second Floor)**



Annex Office





2.1.2 Device Port Assignment Table

Device: Main_RTR1

Port	Connected Device	Port Connected To	Operational Mode
G0/0/0	Main_Dist-SW1	G0/1	Trunk
G0/0/1	Main_Dist-SW2	G0/1	Trunk
S0/1/0	Main_RTR2	S0/1/0	Trunk
S0/1/1	N/A	N/A	Shutdown

Device: Main_RTR2

Port	Connected Device	Port Connected To	Operational Mode
G0/0/0	Main_Dist-SW1	G0/2	Trunk
G0/0/1	Main_Dist-SW2	G0/2	Trunk
S0/1/0	Main_RTR1	S0/1/0	Trunk
S0/1/1	N/A	N/A	Shutdown

Device: Main_Dist-SW1

Port	Connected Device	Port Connected To	Operational Mode
G0/1	Main_RTR1	G0/0/0	Trunk

G0/2	Main_RTR2	G0/0/0	Trunk
Fa0/1	Main_Dist-SW2	Fa0/3	Trunk
Fa0/2	Main_Dist-SW2	Fa0/4	Trunk
Fa0/3	Main_IT-Acc-SW	Fa0/1	Trunk
Fa0/4	Main_Admin-Acc-SW	Fa0/1	Trunk
Fa0/5	Main_Faculty-Acc-SW	Fa0/1	Trunk
Fa0/6	Main_Library-Acc-SW	Fa0/1	Trunk
Fa0/7	Main_TR1-Acc-SW	Fa0/1	Trunk
Fa0/8	Main_TR2-Acc-SW	Fa0/1	Trunk
Fa0/9	Main_TR3-Acc-SW	Fa0/1	Trunk
Fa0/10	Main_TR4-Acc-SW	Fa0/1	Trunk
Fa0/11	Main_TR5-Acc-SW	Fa0/1	Trunk
Fa0/12	Main_TR6-Acc-SW	Fa0/1	Trunk
Fa0/13 - Fa0/23	N/A	N/A	Shutdown
Fa0/24	WLC	G0/1	Trunk

Device: Main_Dist-SW2

Port	Connected Device	Port Connected To	Operational Mode
G0/1	Main_RTR1	G0/0/1	Trunk
G0/2	Main_RTR2	G0/0/1	Trunk
Fa0/1	Main_Dist-SW1	Fa0/3	Trunk
Fa0/2	Main_Dist-SW1	Fa0/4	Trunk
Fa0/3	Main_IT-Acc-SW	Fa0/2	Trunk
Fa0/4	Main_Admin-Acc-SW	Fa0/2	Trunk
Fa0/5	Main_Faculty-Acc-SW	Fa0/2	Trunk
Fa0/6	Main_Library-Acc-SW	Fa0/2	Trunk
Fa0/7	Main_TR1-Acc-SW	Fa0/2	Trunk
Fa0/8	Main_TR2-Acc-SW	Fa0/2	Trunk
Fa0/9	Main_TR3-Acc-SW	Fa0/2	Trunk
Fa0/10	Main_TR4-Acc-SW	Fa0/2	Trunk
Fa0/11	Main_TR5-Acc-SW	Fa0/2	Trunk
Fa0/12	Main_TR6-Acc-SW	Fa0/2	Trunk
Fa0/13 - Fa0/24	N/A	N/A	Shutdown

Device: Main_IT-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	WLC	G0/1	Trunk
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/5	Trunk
Fa0/2	Main_Dist-SW2	Fa0/5	Trunk
Fa0/3	Main_Management-PC	NIC	Access(VLAN 500)
Fa0/4	Orion Server	NIC	Access(VLAN 700)
Fa0/5	Aries Server	NIC	Access(VLAN 500)

Fa0/6	Aquarius Server	NIC	Access(VLAN 700)
Fa0/7 - Fa0/11	Main_IT-PC1 - Main_IT-PC5	NIC	Access(VLAN 600)
Fa0/12 - Fa0/23	N/A	N/A	Shutdown
Fa0/24	First Floor AP	G0	Trunk

Device: Main_Admin-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/6	Trunk
Fa0/2	Main_Dist-SW2	Fa0/6	Trunk
Fa0/3 - Fa0/13	Main_Operations-PC1 - Main_Operations-PC11	NIC	Access(VLAN 400)
Fa0/14 - Fa0/19	N/A	N/A	Shutdown
Fa0/20	Main_Service-Printer	NIC	Access(VLAN 700)
Fa0/21 - Fa0/24	N/A	N/A	Shutdown

Device: Main_Faculty-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/7	Trunk
Fa0/2	Main_Dist-SW2	Fa0/7	Trunk
Fa0/3	Main_Secretary-PC	NIC	Access(VLAN 400)
Fa0/4 - Fa0/13	Main_Instructor-PC1 - Main_Instructor-PC10	NIC	Access(VLAN 300)
Fa0/14 - Fa0/19	N/A	N/A	Shutdown
Fa0/20	Main_Service-Printer2	NIC	Access(VLAN 700)
Fa0/21 - Fa0/23	N/A	N/A	Shutdown
Fa0/24	Second Floor AP	G0	Trunk

Device: Main_Library-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/8	Trunk
Fa0/2	Main_Dist-SW2	Fa0/8	Trunk
Fa0/3	Main_Librarian-PC	NIC	Access(VLAN 400)
Fa0/4 - Fa0/9	Main_Guest-PC1 - Main_Guest-PC6	NIC	Access(VLAN 200)

<i>Fa0/10 - Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
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Device: Main_TR1-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
<i>G0/1</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>G0/2</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>Fa0/1</i>	<i>Main_Dist-SW1</i>	<i>Fa0/9</i>	<i>Trunk</i>
<i>Fa0/2</i>	<i>Main_Dist-SW2</i>	<i>Fa0/9</i>	<i>Trunk</i>
<i>Fa0/3</i>	<i>Main_Instructor-TR1-PC</i>	<i>NIC</i>	<i>Access(VLAN 300)</i>
<i>Fa0/4 - Fa0/23</i>	<i>Main_Student-PC1 - Main_Student-PC20</i>	<i>NIC</i>	<i>Access(VLAN 100)</i>
<i>Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>

Device: Main_TR2-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
<i>G0/1</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>G0/2</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>Fa0/1</i>	<i>Main_Dist-SW1</i>	<i>Fa0/10</i>	<i>Trunk</i>
<i>Fa0/2</i>	<i>Main_Dist-SW2</i>	<i>Fa0/10</i>	<i>Trunk</i>
<i>Fa0/3</i>	<i>Main_Instructor-TR2-PC</i>	<i>NIC</i>	<i>Access(VLAN 300)</i>
<i>Fa0/4 - Fa0/23</i>	<i>Main_Student-PC1 - Main_Student-PC20</i>	<i>NIC</i>	<i>Access(VLAN 100)</i>
<i>Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>

Device: Main_TR3-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
<i>G0/1</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>G0/2</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>Fa0/1</i>	<i>Main_Dist-SW1</i>	<i>Fa0/11</i>	<i>Trunk</i>
<i>Fa0/2</i>	<i>Main_Dist-SW2</i>	<i>Fa0/11</i>	<i>Trunk</i>
<i>Fa0/3</i>	<i>Main_Instructor-TR3-PC</i>	<i>NIC</i>	<i>Access(VLAN 300)</i>
<i>Fa0/4 - Fa0/23</i>	<i>Main_Student-PC1 - Main_Student-PC20</i>	<i>NIC</i>	<i>Access(VLAN 100)</i>
<i>Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>

Device: Main_TR4-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
<i>G0/1</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>G0/2</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>
<i>Fa0/1</i>	<i>Main_Dist-SW1</i>	<i>Fa0/12</i>	<i>Trunk</i>
<i>Fa0/2</i>	<i>Main_Dist-SW2</i>	<i>Fa0/12</i>	<i>Trunk</i>
<i>Fa0/3</i>	<i>Main_Instructor-TR4-PC</i>	<i>NIC</i>	<i>Access(VLAN 300)</i>
<i>Fa0/4 - Fa0/23</i>	<i>Main_Student-PC1 - Main_Student-PC20</i>	<i>NIC</i>	<i>Access(VLAN 100)</i>
<i>Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>

Device: Main_TR5-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/13	Trunk
Fa0/2	Main_Dist-SW2	Fa0/13	Trunk
Fa0/3	Main_Instructor-TR5-PC	NIC	Access(VLAN 300)
Fa0/4 - Fa0/23	Main_Student-PC1 - Main_Student-PC20	NIC	Access(VLAN 100)
Fa0/24	N/A	N/A	Shutdown

Device: Main_TR6-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Main_Dist-SW1	Fa0/14	Trunk
Fa0/2	Main_Dist-SW2	Fa0/14	Trunk
Fa0/3	Main_Instructor-TR6-PC	NIC	Access(VLAN 300)
Fa0/4 - Fa0/23	Main_Student-PC1 - Main_Student-PC20	NIC	Access(VLAN 100)
Fa0/24	N/A	N/A	Shutdown

2.2. Annex**2.2.1. Floor Plan****2.2.2. Device Port Assignment Table****Device: Annex_RTR1**

Port	Connected Device	Port Connected To	Operational Mode
G0/0/0	Annex_Dist-SW1	G0/1	Trunk
G0/0/1	N/A	N/A	Shutdown
S0/1/0	N/A	N/A	Shutdown
S0/1/1	N/A	N/A	Shutdown

Device: Annex_Dist-SW1

Port	Connected Device	Port Connected To	Operational Mode
G0/1	Annex_RTR1	G0/0/0	Trunk
G0/2	N/A	N/A	Shutdown
Fa0/1	Annex_Faculty-Acc-SW	Fa0/1	Trunk
Fa0/2	Annex_IT-Acc-SW	Fa0/1	Trunk
Fa0/3	Annex_TR1-Acc-SW	Fa0/1	Trunk
Fa0/4	Annex_TR2-Acc-SW	Fa0/1	Trunk
Fa0/5	Annex_TR3-Acc-SW	Fa0/1	Trunk
Fa0/6 - Fa0/24	N/A	N/A	Shutdown

Device: Annex_IT-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Annex_Dist-SW1	Fa0/2	Trunk
Fa0/2	N/A	N/A	Shutdown
Fa0/3	N/A	N/A	Shutdown
Fa0/4	N/A	N/A	Shutdown
Fa0/5	Annex_Management PC	NIC	Access(VLAN 600)
Fa0/6	Annex_IT-PC1	NIC	Access(VLAN 600)
Fa0/7	Annex_IT-PC2	NIC	Access(VLAN 300)
Fa0/8	Annex_Instructor-PC1	NIC	Access(VLAN 300)
Fa0/9	Annex_Instructor-PC2	NIC	Access(VLAN 300)
Fa0/11	Annex_Instructor-PC3		
Fa0/10 - Fa0/24	N/A	N/A	Shutdown

Device: Annex_TR1-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Annex_Dist-SW1	Fa0/3	Trunk
Fa0/2	Annex_Instructor-TR1-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	Annex_StudentPC-1 - Annex_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

Device: Annex_TR2-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Annex_Dist-SW1	Fa0/4	Trunk
Fa0/2	Annex_Instructor-TR2-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	Annex_StudentPC-1 - Annex_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

Device: Annex_TR3-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	Annex_Dist-SW1	Fa0/5	Trunk
Fa0/2	Annex_Instructor-TR3-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	Annex_StudentPC-1 - Annex_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

2.3 Quezon City Site

2.3.1 Floor Plan

2.3.2 Device Port Assignment Table

Device: QC_RTR1

Port	Connected Device	Port Connected To	Operational Mode
G0/0/0	QC_Dist-SW1	G0/1	Trunk
G0/0/1	N/A	N/A	Shutdown
S0/1/0	N/A	N/A	Shutdown
S0/1/1	N/A	N/A	Shutdown

Device: QC_Dist-SW1

Port	Connected Device	Port Connected To	Operational Mode
G0/1	QC_RTR1	G0/0/0	Trunk
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Admin-Acc-SW	Fa0/1	Trunk
Fa0/2	QC_Faculty-Acc-SW	Fa0/1	Trunk
Fa0/3	QC_IT-Acc-SW	Fa0/1	Trunk
Fa0/4	QC_TR1-Acc-SW	Fa0/1	Trunk
Fa0/5	QC_TR2-Acc-SW	Fa0/1	Trunk
Fa0/6	QC_TR3-Acc-SW	Fa0/1	Trunk
Fa0/7	QC_TR4-Acc-SW	Fa0/1	Trunk
Fa0/8 - Fa0/24	N/A	N/A	Shutdown

Device: QC_Admin-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/1	Trunk
Fa0/2 - Fa0/11	QC_Operations-PC1 - QC_Operations-PC10	N/A	Access(VLAN 400)
Fa0/12 - Fa0/24	N/A	N/A	Shutdown

Device: QC_Faculty-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/2	Trunk
Fa0/2 - Fa0/5	QC_Instructor-PC1 - QC_Instructor-PC4	NIC	Access(VLAN 300)
Fa0/6 - Fa0/24	N/A	N/A	Shutdown

Device: QC_IT-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown

G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/3	Trunk
Fa0/2	QC_Management-PC	N/A	Access(VLAN 500)
Fa0/3	Leo Server	N/A	Access(VLAN 500)
Fa0/4 - Fa0/9	QC_IT-PC1 - QC_IT-PC6	N/A	Access(VLAN 600)
Fa0/10 - Fa0/24	N/A	N/A	Shutdown

Device: QC_TR1-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/4	Trunk
Fa0/2	QC_Instructor-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	QC_StudentPC-1 - QC_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

Device: QC_TR2-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/5	Trunk
Fa0/2	QC_Instructor-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	QC_StudentPC-1 QC_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

Device: QC_TR3-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/6	Trunk
Fa0/2	QC_Instructor-PC	NIC	Access(VLAN 300)
Fa0/3 - Fa0/22	QC_StudentPC-1 - QC_StudentPC-20	NIC	Access(VLAN 100)
Fa0/23 - Fa0/24	N/A	N/A	Shutdown

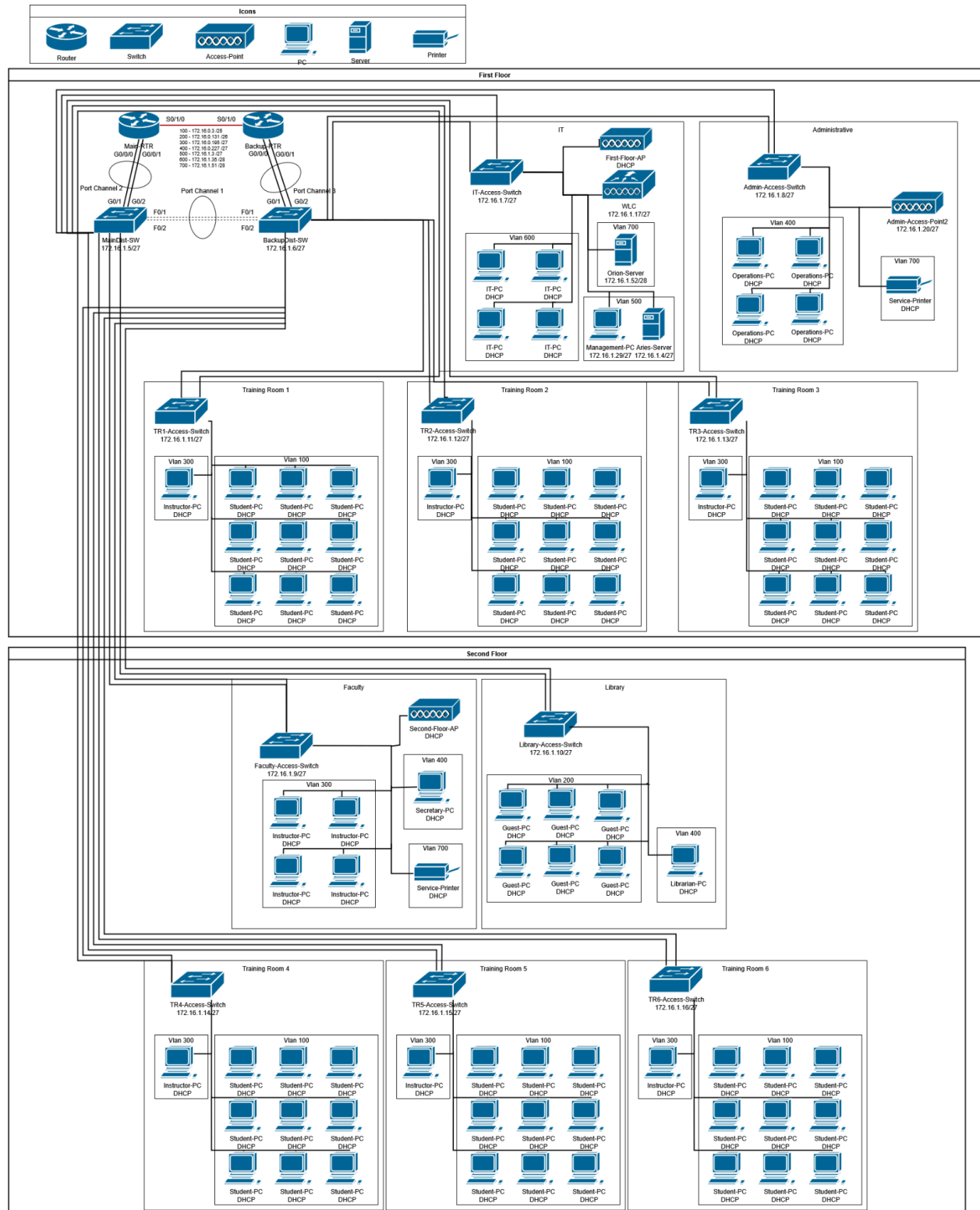
Device: QC_TR4-Acc-SW

Port	Connected Device	Port Connected To	Operational Mode
G0/1	N/A	N/A	Shutdown
G0/2	N/A	N/A	Shutdown
Fa0/1	QC_Dist-SW1	Fa0/7	Trunk
Fa0/2	QC_Instructor-PC	NIC	Access(VLAN 300)
Fa0/3 -	QC_StudentPC-1 -	NIC	Access(VLAN 100)

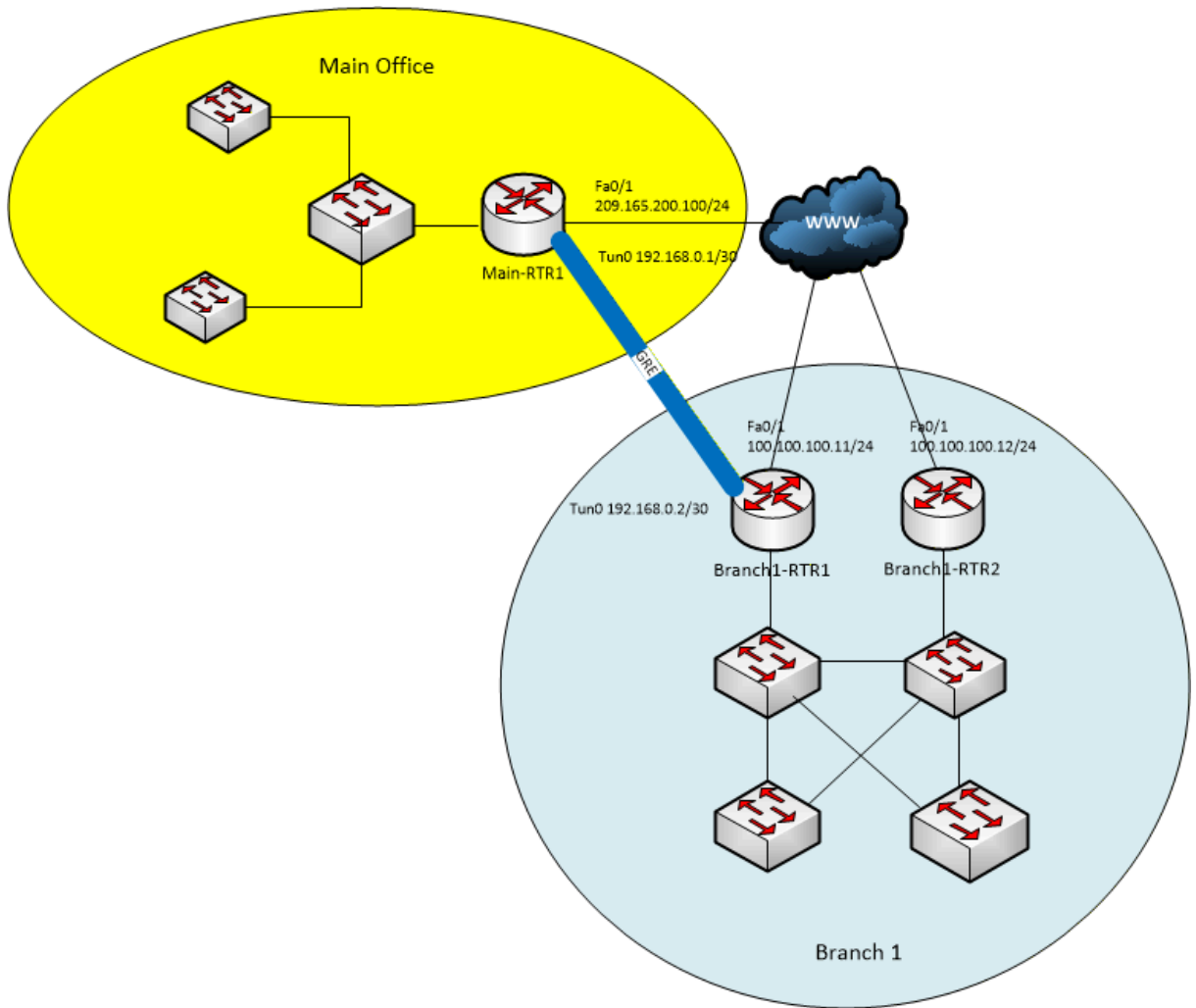
<i>Fa0/22</i>	<i>QC_StudentPC-20</i>		
<i>Fa0/23 - Fa0/24</i>	<i>N/A</i>	<i>N/A</i>	<i>Shutdown</i>

3. **Logical Topology** - Separate image files may be submitted if diagrams are too large for details to be clearly seen in the main document
- **Overall organization diagram** illustrates all sites with their Internet connections and any site-to-site connections
 - **Per site Diagram** provides the topology details of each site's internal network
 - Include ALL infrastructure devices (e.g. routers, switches, WLC, Aps) and devices providing network services (e.g. servers and printers) – labelled with hostnames and IP addresses
 - Include representative end devices per user group – labelled with group names and network addresses

Main Office

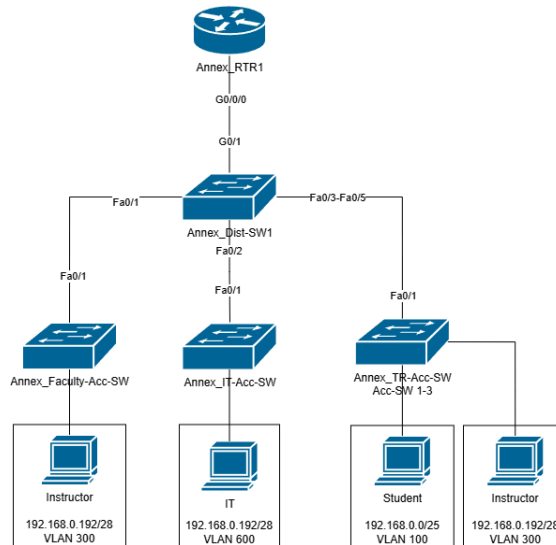


3.1. Organization Network (Needed in Phase 2)

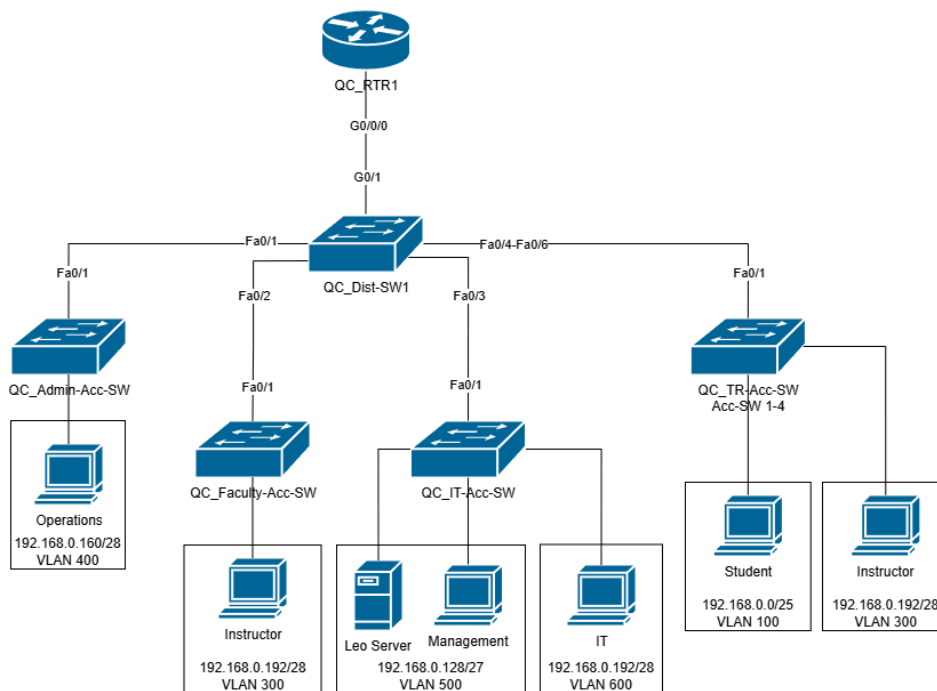


3.2. Manila-Main Office

3.3. Manila-Annex Office



3.4. Quezon City Office



4. Security Implementation

4.1. Configured Credentials

4.1.1 Main Office

Device	Privileged EXEC Password	Console Password	Remote Access Password
Main_RTR1	Gonzalez-7xQ	Nairobi-zu!s	User: main-rtr-user Password: Ecosport-ika!
Main_RTR2	Garcia-3jNzo	Baguio-1aMig	User: backup-rtr-user Password: Escape-oOpG
Main_Dist-SW1	torreS-Th9!	istanbuL-7rk!	User: main-sw-user Password:taurus-sN1c
Main_Dist-SW2	ortegA-k4a3	hamburG-G3r	User: backup-sw-user Password: focuS-a66w
Main_IT-Acc-SW	alvareZ-iW3	viennA-faRR	User: it-sw-user Password: broncO-\$imbz
Main_Admin-Acc-SW	mendozA-G9k	kyotO-j!os	User: admin-sw-user Password: mustanG-3nRg
Main_Faculty-Acc-SW	fuentes-An4	vermonT-D1a	User: faculty-sw-user Password: rangeR-fm#!
Main_Library-Acc-SW	cabrerA-!!z3	cairO-23E#	User: library-sw-user Password: fusioN-pk1s
Main_TR1-Acc-SW	RiverA-0os6	WarsaW-!!sz	User: tr1-sw-user

			Password: LanceR-3v0x
Main_TR2-Acc-SW	MoraleS-sz!o	Dubal-exP1	User: tr2-sw-user Password: ExploreR-21m.
Main_TR3-Acc-SW	SalazaR-Wk\$	LagoS-ok29	User: tr3-sw-user Password: AvaloN-h!r3
Main_TR4-Acc-SW	MartineZ-782	Seville-StRt	User: tr4-sw-user Password: InnovA-1hhA
Main_TR5-Acc-SW	Castillo-J3L	DavaO-jK31	User: tr5-sw-user Password: CrosS-m#iz0
Main_TR6-Acc-SW	FloreS-p1u	BostoN-#wa	User: tr6-sw-user Password: CoopeR-mTOR

Device	VTP Mode	VTP Domain	VTP Password
Main_Dist-SW1	Server	AITC-NET.com	a1onz0-vtp-p@ss
Main_Dist-SW2	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_IT-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_Admin-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_Faculty-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_Library-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR1-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR2-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR3-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR4-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR5-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Main_TR6-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss

Device	SNMP Community Password Management (READ AND WRITE)	SNMP Community Password IT (READ ONLY)
Main_RTR1	Nebula-ip4!	Quasar-ip4!
Main_RTR2	Comet-mar#	Umbra-mar#
Main_Dist-SW1	novA-r00a	oorT-r00a

Main_Dist-SW2	saturN-jn1d	bolidE-jn1d
Main_IT-Acc-SW	venuS-09ik	coronaL-09ik
Main_Admin-Acc-SW	jupiteR-7jm	rochE-7jm
Main_Faculty-Acc-SW	kepleR-pi3	mooN-pi3
Main_Library-Acc-SW	pistoL-4!z	voiD-4!z
Main_TR1-Acc-SW	PolariS-Şoar	QuarK-Şoar
Main_TR2-Acc-SW	AsteroiD-&ij	VestA-&ij
Main_TR3-Acc-SW	SputniK-ak3	CereS-ak3
Main_TR4-Acc-SW	EuropA-gu3	AsteR-gu3
Main_TR5-Acc-SW	TritoN-naw2	TitaN-naw2
Main_TR6-Acc-SW	PhoboS-bisk	SolaR-bisk

4.1.2 Annex Office

Device	Privileged EXEC Password	Console Password	Remote Access Password
Annex_RTR1	Reyes-9aP#	Tokyo-p9q!	User: annex-rtr-user Password: Civic-82xA
Annex_Dist-SW1	Santos-2bX!	Paris-L4w\$	User: annex-sw-user Password: Corolla-9vM!
Annex_IT-Acc-SW	Cruz-5cZ@	London-x1Y*	User: it-sw-user Password: Accord-3jK#
Annex_Faculty-Acc-SW	Bautista-8dW%	Berlin-2aC@	User: faculty-sw-user Password: Camry-7hL!
Annex_TR1-Acc-SW	Ocampo-1eY^	Madrid-b3V#	User: tr1-sw-user Password: Sentra-5gN@
Annex_TR2-Acc-SW	Aquino-4fV&	Rome-c8B!	User: tr2-sw-user Password: Altima-2fP\$
Annex_TR3-Acc-SW	Ramos-7gU*	Athens-m5N%	User: tr3-sw-user Password: Mazda3-4dC^

Device	VTP Mode	VTP Domain	VTP Password
Annex_Dist-SW1	Server	AITC-NET.com	a1onz0-vtp-p@ss
Annex_IT-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Annex_Faculty-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Annex_TR1-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Annex_TR2-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
Annex_TR3-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss

Device	SNMP Community Password Management (READ AND WRITE)	SNMP Community Password IT (READ ONLY)
Annex_RTR1	Galaxy-9aP#	Pulsar-9aP#
Annex_Dist-SW1	Meteor-2bX!	Orbit-2bX!
Annex_IT-Acc-SW	Zenith-5cZ@	Nadir-5cZ@
Annex_Faculty-Acc-SW	Planet-8dW%	Star-8dW%
Annex_TR1-Acc-SW	Cosmos-1eY^	Void-1eY^
Annex_TR2-Acc-SW	Aurora-4fV&	Halo-4fV&
Annex_TR3-Acc-SW	Eclipse-7gU*	Lunar-7gU*

4.1.3 QC

Device	Privileged EXEC Password	Console Password	Remote Access Password
QC_RTR1	Gomez-2wQ!	Seoul-7vB#	User: qc-rtr-user Password: Tucson-1xZ@
QC_Dist-SW1	Lopez-5eR@	Taipei-3nC\$	User: qc-sw-user Password: Sonata-8yW!
QC_IT-Acc-SW	Navarro-8tY#	Manila-9mX%	User: it-sw-user Password: Elantra-4vS#

QC_Faculty-Acc-SW	Castro-3uI\$	Bangkok-4zL^	User: faculty-sw-user Password: Rio-6bN!
QC_Admin-Acc-SW	Diaz-6oP%	Hanoi-1kK&	User: admin-sw-user Password: Forte-9cV@
QC_TR1-Acc-SW	Perez-9sA^	Jakarta-6jJ*	User: tr1-sw-user Password: Optima-2mQ\$
QC_TR2-Acc-SW	Ruiz-1dF&	Mumbai-8hH!	User: tr2-sw-user Password: Soul-5nE%
QC_TR3-Acc-SW	Sosa-4gG*	Dubai-2fF@	User: tr3-sw-user Password: Spark-7rT^
QC_TR4-Acc-SW	Vega-7hJ!	Doha-5dD#	User: tr4-sw-user Password: Sonic-3pY&

Device	VTP Mode	VTP Domain	VTP Password
QC_Dist-SW1	Server	AITC-NET.com	a1onz0-vtp-p@ss
QC_IT-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_Faculty-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_Admin-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_TR1-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_TR2-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_TR3-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss
QC_TR4-Acc-SW	Client	AITC-NET.com	a1onz0-vtp-p@ss

Device	SNMP Community Password Management (READ AND WRITE)	SNMP Community Password IT (READ ONLY)
QC_RTR1	Orion-2wQ!	Lyra-2wQ!
QC_Dist-SW1	Cygnus-5eR@	Draco-5eR@
QC_IT-Acc-SW	Vela-8tY#	Ara-8tY#
QC_Faculty-Acc-SW	Leo-3uI\$	Aries-3uI\$
QC_Admin-Acc-SW	Taurus-6oP%	Gemini-6oP%

QC_TR1-Acc-SW	Virgo-9sA^	Libra-9sA^
QC_TR2-Acc-SW	Ursa-1dF&	Canis-1dF&
QC_TR3-Acc-SW	Hydra-4gG*	Lynx-4gG*
QC_TR4-Acc-SW	Pavo-7hJ!	Grus-7hJ!

4.2. Access Control Matrix

*Include only intra-site and Internet access permissions for phase 1. Add entries for inter-site access permissions in phase 2

* A separate file / spreadsheet may be used for this section if the matrix cannot easily fit into the main document

* This may also be created as single large matrix containing all groups across all sites

ITNET04 Access Control List Matrix

4.2.1 Main Office

		Receiver						Management
		Students	Guests	Instructors	Operations	Mgmt	IT	
Sender	Main-Staff		X		IP	X		
	Main-Officers	X			x	IP		
	...							
	(add rows as needed)							

4.2.2 Branch 1

		Receiver					
		Branch1-Staff	Branch1-Officers	Group 3 name	Main-Staff	Main-Officers	(add columns as needed)
Sender	Branch1-Staff		X		IP	X	
	Branch1-Officers	X			X	IP	
	...						
	(add rows as needed)						

5. Design Discussion

Write a technical discussion of your design in paragraph form. Your discussion should generally answer the following guide questions:

- What technology / design was implemented to address a requirement?
- How was the technology / protocol / design implemented (i.e. configuration options and parameters)?
- Why was the technology / protocol / design selected and why did you implement it in the way that you did?
- What alternative approaches were considered, and why were they not selected?

Sample:

To support communication between the Main Office and Branch1, static routing with route summarization was implemented. The Main Office uses the address space of 192.168.0.0/22, while the Branch 1 was allocated subnets from 192.168.4.0/24. Static routes were configured on the site routers, with the Main router maintaining summarized routes for the Branch1 site through their respective VPN tunnels. Likewise, the Branch1 router use a default route toward the Main site for organizational resources and Internet access. This design reduces the number of routing table entries because each branch can be represented by a single summary network rather than multiple VLAN subnets. Static routing was selected because the network consists of only two sites with a predictable topology, making it simpler to configure and troubleshoot than a dynamic routing protocol. OSPF was considered as an alternative because it would automatically learn routes and adapt to topology changes; however, the additional complexity and protocol overhead were not justified for a network of this size.

6. AI Use Disclosure (if applicable)

6.1. AI Use Reflection

- Which tool(s) were used
- Purpose(s) of use
- How AI output differed from own idea
- How correctness was verified
- What was learned

Sample:

For this project, I used ChatGPT in two specific ways:

- **Concept clarification:** *I asked for a step-by-step explanation of Variable Length Subnet Masking (VLSM) because I was unsure how to allocate subnets efficiently for different network sizes. This helped me review the concept, but I created the actual IP addressing plan on my own.*
- **Documentation support:** *After writing my network design documentation, I used AI to suggest improvements for clarity and grammar. I then revised the document based on these suggestions.*

I verified the correctness of my work by recalculating all subnets manually and testing my addressing plan in Cisco Packet Tracer. I also cross-checked the AI's grammar suggestions against my original text to ensure that technical meaning was not changed.

From this process, I learned how to better apply VLSM to different subnet requirements and how to present my documentation in a more professional style.

6.2. AI Prompts Used

- Provide a copy of the exact prompts/inputs submitted to the AI tool.

Sample:

1. Concept Clarification (VLSM)

- *"How do I decide subnet sizes when different departments have different host requirements?"*
- *"Show me the step-by-step process for subnetting a 192.168.1.0/24 network using VLSM if one subnet needs 50 hosts, another 25 hosts, and two others need 10 hosts each."*

2. Documentation Support (Polishing Report)

- *"Please check the grammar and readability of this section about my IP addressing scheme."*
- *"Can you format this documentation section so that it's clearer for technical readers?"*